

Account Aggregator Platform - Evaluator Manual

Account Aggregator Platform

Project Overview

This project implements a secure Account Aggregator platform that enables consent-based retrieval of customer financial information.

Technology Stack

Backend: Java 17, Spring Boot, Spring Security, Spring Data JPA

Database: PostgreSQL

AI: GenAI API Integration Ready

Project Structure

pom.xml - Maven dependency management

application.properties - Application configuration

AaBackendApplication.java - Main application entry point

Authentication Module

User.java - User entity

UserRepository.java - User database access

RegisterRequest.java - Registration DTO

LoginRequest.java - Login DTO

SecurityConfig.java - BCrypt configuration

AuthService.java - Authentication logic

AuthController.java - Authentication APIs

Consent Module

Consent.java - Consent entity

ConsentRepository.java - Consent persistence

CreateConsentRequest.java - Consent DTO

ConsentService.java - Consent business logic

ConsentController.java - Consent API

Financial Data Module

Account.java - Account entity

AccountRepository.java - Account persistence

Transaction.java - Transaction entity

TransactionRepository.java - Transaction persistence

AI Module

GenAIService.java - Financial summary generation

Dashboard Module

DashboardController.java - Dashboard API

Security Features

- BCrypt Password Hashing
- Session-Based Authentication
- 30 Minute Session Timeout
- Secure Cookie Support
- API Key Protection

How to Run

1. Install Java 17, Maven and PostgreSQL.

2. Create database:

```
CREATE DATABASE aa_db;
```

3. Update application.properties with PostgreSQL credentials.

4. Open terminal:

```
cd aa-backend
```

5. Build:

```
mvn clean install
```

6. Run:

```
mvn spring-boot:run
```

7. Open:

```
http://localhost:8080
```

Test APIs:

```
POST /api/auth/register
```

```
POST /api/auth/login
```

```
POST /api/consents
```

```
GET /api/dashboard/summary
```

Future Enhancements

- Full Digio Integration
- Transaction Categorization
- Credit Risk Analysis
- Spending Analytics